F24-134-D-TreeVerse

Project Team

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Chapter 1

Introduction

TreeVerse is a Global Tree Mapping and Monitoring System that aims to leverage machine learning techniques, utilizing Google Maps data, and satellite imagery to develop a com- prehensive platform for identifying and monitoring trees across the globe. Furthermore, the system will also analyze forestation and deforestation trends and provide actionable insights for sustainable plantation efforts.

1.1 Problem Statement

We are developing a Global Tree Mapping and Monitoring system for our final-year project. The reason is that it solves problems we face today related to forests. First of all, tree classification is a major problem in deep forests. While we can easily identify plants that are easily accessible using our smartphones, deep forests present a different set of challenges. Like accessibility, danger, and lack of proper paths. We plan to identify tree types in these inaccessible areas. Additionally, in recent years we have seen unprecedented deforestation around the globe, making it difficult to track these trends and create effective policies. Lastly, people and Non-Governmental Organizations have realized the importance of plantations in curbing the adverse effects of climate change. This has created a very unique problem of where to plant new plants and which plants will be suitable for that specific region. we are planning to solve these problems in our project.

1.2 Scope

Our project TreeVerse is a project aimed at creating a system that can successfully map different types of vegetation around the world, and identify the deforestation trends in an area. The geographic scope of our project will initially cover a small area of Mansehra. In this area, we will identify the different types of vegetation and the trends of deforestation over the years. Regarding the identification of the types of vegetation, we will differentiate between trees, bushes, barren land, and crops. We will monitor and frame the different types of vegetation on a map with the help of satellite and geospatial imagery. We will obtain our satellite imagery from our external collaborators, Google Earth Engine, Google Maps API, and PlanetLabs. This data will feature multi-spectral data which will be essential for the Machine Learning algorithms to be used in our project. The Machine Learning algorithms we will employ range greatly according to their application in our project. As an example, for the purpose of tree counting and mapping, we will employ tree counting algorithms like Random Forest, Dark Forest, etc. These machine-learning algorithms will help us monitor deforestation trends and map the vegetation types by clustering, counting, classification, and other methodologies. We hope to effectively complete our project at least up to the definition in the above-mentioned scope.

1.3 Modules

TreeVerse is organized into the following modules, each focusing on key functionalities required to deliver a comprehensive platform.

1.3.1 Data Processing

This module handles the gathering, storing, and preprocessing of satellite imagery. The goal is to ensure that the data is clean, normalized, and ready for algorithm designing, model training, and integration with cloud storage for data management.

1. Satellite imagery gathering and storage
2. Data preprocessing and cleaning
3. Image enhancement and normalization
4. Integration with Google Cloud Storage (or Amazon S3)

1.3.2 Tree Classification

At the core of the system's analytical capabilities, this module implements deep learning models for tree species identification. It manages model training, validation, and integration with GPU acceleration services for optimal performance.

1. Trees classification algorithms

2. Implementation of deep learning models (PyTorch)
3. Model training and validation

1.3.3 Deforestation Monitoring

This module focuses on detecting and analyzing changes in forest cover over time. It implements algorithms for change detection, generates alerts for significant deforestation events, and integrates climate data for contextual analysis.

1. Change detection algorithms
2. Time series analysis of forest cover
3. Implementation of deep learning models (PyTorch)
4. Model training and validation
5. Alert generation for significant changes

1.3.4 API Development

Serving as the bridge between the back-end and front-end, this module provides RESTful API endpoints for data access. It handles authentication, authorization, and efficient data querying to support the system's various interfaces.

1. RESTful API endpoints for data access
2. Authentication and authorization
3. Data querying and filtering capabilities
4. Integration with front-end systems

1.3.5 Client Side Web App

The Client Side Web App is a responsive, user-centric interface built with React and Tailwind CSS, offering seamless user authentication and personalized dashboards. It features interactive map displays of forest cover, dynamic charts for trend analysis, and advanced search functionality for specific regions or tree categories. The app provides an intuitive platform for users to access key metrics, receive alerts, and manage their preferences, all within a visually appealing and efficient design.

1. Responsive design using Tailwind CSS
2. User authentication and profile management
3. Dashboard for key metrics and alerts
4. User Preferences and settings
5. Interactive map display of forest cover
6. Dynamic charts and graphs for forest trends
7. Search functionality for specific regions or tree category

1.3.6 Admin Side Web App

The Admin Side Web App is a powerful management interface designed for system administrators and data stewards. It provides comprehensive tools for user account management, including role-based access control and detailed activity logging. The app features robust data handling capabilities, offering manual import/export tools and data quality monitoring. Additionally, it includes advanced functionalities for ML model performance monitoring, enabling administrators to maintain and optimize the system's analytical capabilities.

1. User account creation and management
2. Role-based access control
3. Activity logging and auditing
4. Manual data import and export tools
5. Data quality monitoring and correction tools
6. ML model performance monitoring

1.3.7 Documentation

The Documentation module provides comprehensive resources for system usage and integration. It includes user manuals for all roles and detailed API documentation, ensuring effective utilization and extensibility of the Global Tree Mapping and Monitoring System.

1. User manuals and guides
2. API documentation

1.4 User Classes and Characteristics

Following are user classes that will use TreeVerse.

User class	Description
General User	A General User is a person who uses TreeVerse for general purpose without focusing on a specific goal. He/she may use TreeVerse just to inform themselves about the world of forests and trees. They may use TreeVerse for basic reports and Visualizations
Admin	The admin is the user who is responsible for the correct functioning of TreeVerse. He/She is responsible for managing the data, visualization options, account creation, account deletion, account management, access management, and monitoring activities being done on the system. He/She is also responsible for managing the upkeep of the deployed software.
Government Agency / Environmental Agency	Government Agencies are also one of our main users. They will access TreeVerse to learn and gather information on the areas with increased deforestation in recent times and track the different types of vegetation in an area using TreeVerse's advanced visualization and reporting tools.

Chapter 2

Project Requirements

This chapter describes the functional and non-functional requirements of the project.

2.1 Use-case/Event Response Table/Storyboarding

2.1.1 Use-case 1: Data Preprocessing and Management

Use Case ID	UC1
Use Case	Data Preprocessing and Management
Primary Actor	System
Goal	Gather, clean, and preprocess satellite imagery for classification
Preconditions	Raw satellite imagery is available.
Postconditions	Preprocessed images are stored and ready for analysis.
Main Success Scenario	1. System gathers satellite images. 2. System cleans images (noise reduction, normalization). 3. System enhances images for analysis (edge detection, contrast). 4. System validates image quality. 5. Processed images are stored in the database.
Extensions	E1: Image fails validation, Image needs manual review.

Table 2.1: Use-case 1: Data Preprocessing and Management

2.1.2 Use-case 2: Tree Classification from Satellite Imagery

Use Case ID:	UC2
Use Case:	Tree Classification from Satellite Imagery
Primary Actor:	General User, Researcher, Admin
Stakeholders and Interests:	General User: Understand vegetation types. Researcher: Needs detailed classification for research. Admin: Ensures system performance and data accuracy. Environmental Organizations: Interested in monitoring vegetation.
Preconditions:	The user is authenticated. The system has access to satellite imagery. Trained classification model is available.
Postconditions:	The image is classified and results are displayed. Classified data is stored for future use.
Main Success Scenario:	1. User logs in. 2. The user uploads/selects a satellite image. 3. System preprocesses the image (resizes, normalizes). 4. Classification model is applied. 5. Results are displayed (trees, bushes, grass). 6. Data is stored for future reference.
Extensions:	1a: If not authenticated, prompt login. 3a: Invalid image, request a valid image. 4a: Classification fails, show error. 6a: Data storage fails, retry or notify admin.
Special Requirements:	Image processing must be completed within 5 minutes. Classification accuracy must be at least 80%.

Table 2.2: Use-case 2: Tree Classification from Satellite Imagery

2.1.3 Use-case 3: Deforestation Monitoring

Use Case ID:	UC3
Use Case:	Deforestation Monitoring
Primary Actor:	Researcher, Government Agency, Admin
Stakeholders and Interests:	Researcher: Needs historical forest cover data for analysis. Government Agency: Interested in deforestation trends. Admin: Monitors system functionality.
Preconditions:	Historical and current satellite images are available. The deforestation detection model is trained.
Postconditions:	System identifies and highlights deforested areas. Deforestation alerts are sent to subscribed users.
Main Success Scenario:	1. User logs in. 2. The user selects a region and time range for monitoring. 3. The system compares historical and current images. 4. The system detects changes in forest cover. 5. Detected deforestation areas are displayed. 6. Alerts are sent if significant changes are found.
Extensions:	3a: If images are unavailable, notify the user and prompt for new input. 4a: Model fails to detect deforestation, prompt error.
Special Requirements:	Image comparison must be completed within 5 minutes. Accuracy of detection must be at least 80%.

Table 2.3: Use-case 3: Deforestation Monitoring

2.1.4 Use-case 4: Reforestation Suggestion

Use Case ID:	UC4
Use Case:	Reforestation Suggestion
Primary Actor:	Environmental Organization, General User, Government Agency
Stakeholders and Interests:	Environmental Organizations: Wants suggestions for suitable reforestation areas. Government Agencies: Interested in reforestation efforts for policy planning.
Preconditions:	The user provides location and deforested area data. The system has access to vegetation data.
Postconditions:	The system suggests suitable plant species for reforestation.
Main Success Scenario:	1. User logs in. 2. User inputs deforested area details. 3. The system analyzes soil and climate data for the region. 4. The system suggests plant species suitable for reforestation. 5. The user views and downloads the suggestions.
Extensions:	2a: If no data for the region, notify the user. 4a: If the analysis fails, show an error and log issue.
Special Requirements:	The system must generate suggestions within 5 minutes. Accuracy in recommendations must be at least 80%.

Table 2.4: Use-case 4: Reforestation Suggestion

2.1.5 Use-case 5: User Registration and Authentication

Use Case ID:	UC5
Use Case:	User Registration and Authentication
Primary Actor:	General User, Admin
Stakeholders and Interests:	General User: Wants secure access to the system. Admin: Ensures proper user management and security.
Preconditions:	The user is not yet registered or logged in. The user must provide valid registration details.
Postconditions:	The user is registered and authenticated successfully.
Main Success Scenario:	1. The user visits the TreeVerse platform. 2. The user selects the registration option. 3. The user provides details (name, email, etc.). 4. The system validates the information. 5. User creates a password. 6. The user receives a confirmation email. 7. User logs in with credentials.
Extensions:	3a: If registration details are incomplete, prompt for completion. 4a: If validation fails, notify the user. 6a: Email delivery fails, retry or notify support.
Special Requirements:	The user registration process must be completed within 2 minutes. Passwords must adhere to security standards.

Table 2.5: Use-case 5: User Registration and Authentication

2.1.6 Use-case 6: Interactive Map for Tree Data Visualization

Use Case ID:	UC6
Use Case:	Interactive Map for Tree Data Visualization
Primary Actor:	General User, Researcher
Stakeholders and Interests:	General User: Wants to view tree classification and deforestation data visually. Researcher: Needs spatial representation of tree data.
Preconditions:	The user is authenticated. The system has access to classified tree data.
Postconditions:	The user can interact with the map and visualize tree data.
Main Success Scenario:	1. User logs in. 2. The user selects the interactive map feature. 3. The system loads tree classification data on the map. 4. User views tree locations and vegetation types. 5. The user zooms in/out and selects regions for more details.
Extensions:	3a: If map data fails to load, retry or show an error. 5a: If user interaction with the map fails, provide feedback.
Special Requirements:	The map must load within 10 seconds. Zoom and interaction should be smooth and responsive.

Table 2.6: Use-case 6: Interactive Map for Tree Data Visualization

2.1.7 Use-case 7: Admin Data Management

Use Case ID:	UC7
Use Case:	Admin Data Management
Primary Actor:	Admin
Stakeholders and Interests:	Admin: Manages user accounts, data import/export, and system performance.
Preconditions:	The admin has appropriate permission to manage the system.
Postconditions:	The admin successfully manages user accounts and system data.
Main Success Scenario:	1. Admin logs in. 2. Admin navigates to the data management section. 3. Admin imports/exports satellite imagery data. 4. Admin monitors data quality and resolves any issues. 5. Admin manages user accounts, including role assignments.
Extensions:	3a: Data import/export fails, prompt retry or notify the admin. 5a: User account management issues, show error and log it.
Special Requirements:	Admin actions must be logged for audit purposes. Data import/export should be completed within 10 minutes for large datasets.

Table 2.7: Use-case 7: Admin Data Management

The selection of the requirement-gathering technique(s) will depend on the type of project. For instance, • Use case (use case diagram + detail use case) is an effective technique for interactive end-user applications. • Event-response table is for a real-time system in which most of the functionalities are performed at the backend. • Storyboarding for graphically intensive applications.
Create a suitable representation here.

2.2 Functional Requirements

This section describes the functional requirements of the system expressed in the natural language style. This section is typically organized by feature as a system feature name and specific functional requirements associated with this feature. It is just one possible way to arrange them. Other organizational options include arranging functional requirements by use case, process flow, mode of operation, user class, stimulus, and response depending on what kind of technique has been used to understand functional requirements. Hierarchical combinations of these elements are also possible, such as use cases within user classes.

2.2.1 Module 1: Data Processing

The following are the requirements for Module 1:

1. FR1: The system shall contain images from different sources and are securely stored in a database
2. FR2: The system shall preprocess the images and standardize them by removing noise.
3. FR3: The system shall integrate cloud storage services to store and manage satellite images.

2.2.2 Module 2: Tree Classification

The following are the requirements for module 2:

1. FR1: The system shall classify plant species into three categories: bushes, trees, and crops, based on satellite imagery.
2. FR2: The system shall use different algorithms to classify the plants based on satellite images.
3. FR3: The system shall train the classification model on labeled satellite images and validate against a separate data set.

2.2.3 Module 3: Deforestation Monitoring

The following are the requirements for module 3:

1. FR1: The system shall identify changes in forest covers using change identification algorithms over time.
2. FR2: The system shall perform a time series analysis to keep track of deforestation trends.
3. FR3: The system shall validate the deforestation detection model using historical data.

2.2.4 Module 4: API Development

The following are the requirements for module 4:

1. FR1: The system shall provide RESTful API endpoints to the user to interact with the data.

2. FR2: The system shall implement authentication and authorization for secure access to the endpoints.

3. FR3: The system shall integrate properly with the front end for proper data communication.

2.2.5 Module 5: Client Side Web App

The following are the requirements for module 5:

1. FR1: The system shall support user authentication and management.

2. FR2: The system shall display important information on the dashboard of the user.

2.2.6 Module 6:Admin Side Web App

The following are the requirements for module 6:

1. FR1:The system shall allow administrators to create, manage, and delete user accounts, including assigning roles and permissions for role-based access control.

2. FR2: The system shall assign special roles to specific users for controlled data access.

3. FR3: The system shall allow user action monitoring and auditing.

2.3 Non-Functional Requirements

Following are the non-functional requirement that we hope to implement in TreeVerse. These non-functional requirements are essential for a good User Experience and User Interface.

2.3.1 Reliability

The following are the non-functional requirements for reliability of the system:

2.3.1.1 NFR1:

The system should have Mean Time Between Failure **(MTBF)** of at least 30 days for the system to have an uninterrupted operation.

2.3.1.2 NFR2:

If the system fails the system will automatically start normal operation within 15 mins.

2.3.1.3 NFR3:

The system shall have a logging mechanism for failures and errors. They should be visible to the administrator within **5 minutes** of the failure.

2.3.2 Usability

In terms of usability following can be the non-functional requirements:

2.3.2.1 NFR1:

Visualizations for the Classification aspect of TreeVerse and The deforestation trends should be available within 5 clicks after successful login into the system.

2.3.2.2 NFR2:

Visual feedback shall be given for all user actions within a 10-second timeframe.

2.3.3 Performance

When it comes to performance, following requirements must be considered:

2.3.3.1 NFR1:

Visualizations should be displayed for any location queried for within a 15 seconda time-frame.

2.3.3.2 NFR2:

At the minimum 100 concurrent users must be able to access and query the system.

2.3.4 Security

As adhering to security standards following non-functional requirements must be deeply considered:

2.3.4.1 NFR1:

All system data must be inaccessible to non-admin users.

2.3.4.2 NFR2:

All sensitive data in TreeVerse should be encrypted using the AES-256 Standard.

2.4 Domain Model

Domain model representation for TreeVerse.

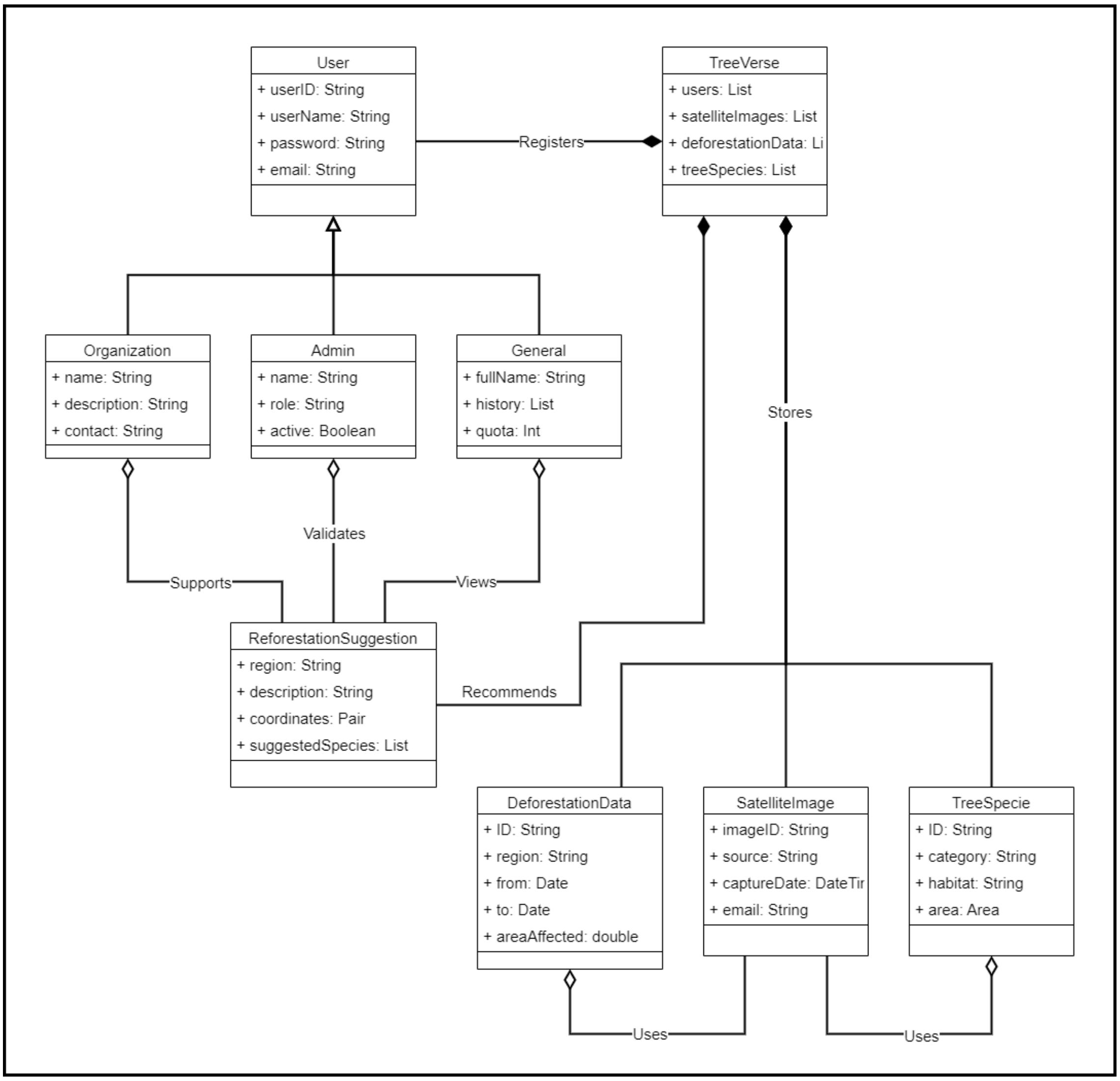

Figure 2.1: Domain model of TreeVerse

Chapter 3

System Overview

TreeVerse is an innovative platform that helps track and protect our world's forests using cutting-edge satellite technology. By combining satellite imagery with advanced mapping tools, we provide a comprehensive view of global forest changes, helping conservationists, researchers, and decision-makers make informed choices about forest management.

What TreeVerse Offers:

1. Real-Time Forest Monitoring Using state-of-the-art satellite data, TreeVerse tracks forest coverage worldwide, distinguishing between trees, shrubs, and grasslands. This detailed mapping helps identify areas of concern and opportunities for conservation.

2. Deforestation Alert System Our platform quickly detects and reports forest loss, providing crucial details about affected regions and potential causes. This timely information enables rapid response to forest threats.

3. Smart Reforestation Planning When forests are lost, TreeVerse doesn't just report the problem—it offers solutions. Our system suggests region-appropriate tree species for replanting efforts, maximizing the success of reforestation projects.

4. User-Friendly Interactive Maps Explore global forest trends through our intuitive mapping interface. Whether you're a researcher analyzing data or a concerned citizen learning about forest conservation, TreeVerse makes complex information accessible and actionable.

5. Tailored Access for Different Users From environmental organizations to government agencies, TreeVerse adapts to various user needs. Each user type gets customized tools and data access, ensuring everyone can contribute effectively to forest conservation efforts.

TreeVerse is more than just a monitoring system—it's a platform for positive change, empowering users worldwide to understand and protect our vital forest ecosystems.

3.1 Architectural Design

Develop a modular program structure and explain the relationships between the modules to achieve the complete functionality of the system. This is a high-level overview of how the system's modules collaborate with each other in order to achieve the desired functionality.

Don't go into too much detail about the individual subsystems. The main purpose is to gain a general understanding of how and why the system was decomposed, and how the individual parts work together.

Provide a diagram showing the major subsystems and their connections.

3.1.1 Box and Line Diagram

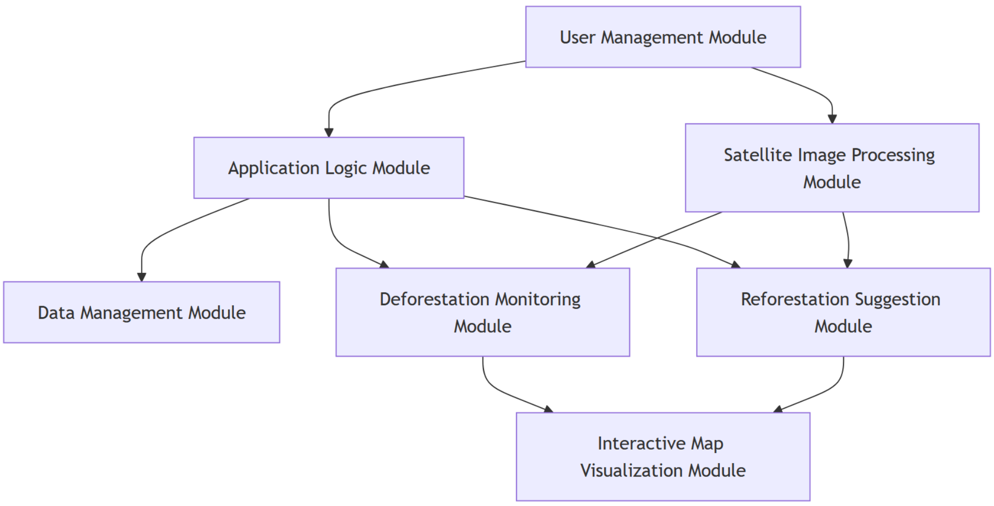

Figure 3.1: Box and Line Diagram

3.1.2 Layered Diagram

3.2 Design Models

Create design models as are applicable to your system. Provide detailed descriptions with each of the models that you add. Also ensure visibility of all diagrams.

3.2.1 Activity Diagram

Design Models for Object-Oriented Development Approach

The applicable models for the project using object-oriented development approach may include:

• Activity Diagram
• Class Diagram
• Class-level Sequence Diagram
• State Transition Diagram (for the projects which include event handling and backend processes)

Design Models for Procedural Approach

The applicable models for the project using a procedural approach may include:

• Activity Diagram
• Data Flow Diagram (data flow diagram should be extended to 2-3 levels. It should clearly list all processes, their sources/sinks and data stores.)
• System-level Sequence Diagram
• State Transition Diagram (for the projects which include event handling and backend processes)

3.3 Data Design

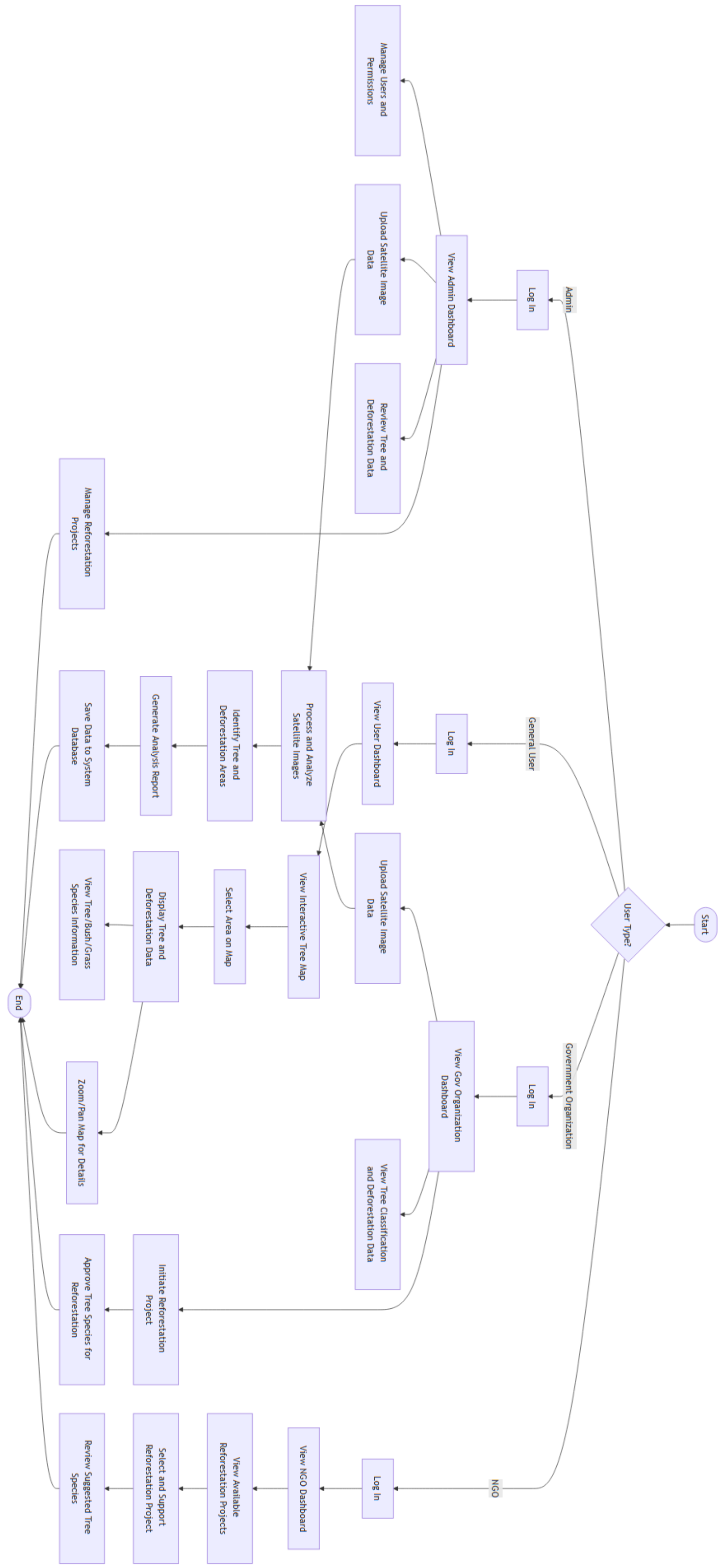

Figure 3.2: Activity Diagram

Bibliography